

Supplementary material for ‘Noisy calibration, latent coverage and the location of the mean’

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Equation, theorem and section numbers without the prefix S refer to the main paper. Numbers that are not proved are labelled as numerical, and the precision of each computation is stated.

S1 Proofs omitted from the main paper

Proof of Proposition 1. Since $\mathcal{F} \subset \mathcal{L}$ it suffices to bound the supremum over $\mathcal{L}$. Let $W \in \mathcal{L}$ satisfy $E\{g_x(W)\} \geq p$ and let $s < Q_q(|W|)$, so that $\text{pr}(|W| < s) \leq q - \gamma$ for some $\gamma > 0$. Choose $M \geq s$ with $g_x(w) \leq p$ for $|w| \geq M$, and let W_M be W conditioned on $|W| \leq M$, which is log-concave. With $\varepsilon_M = \text{pr}(|W| > M)$, the conditioning removes only mass on which the kernel is at most p , so

$$E\{g_x(W_M)\} \geq \frac{p - p\varepsilon_M}{1 - \varepsilon_M} = p, \quad \text{pr}(|W_M| < s) \leq \frac{q - \gamma}{1 - \varepsilon_M} < q$$

for M large. On $K = [-M, M]$ the constraint function $g_x - p$ is continuous, and $\mu \mapsto -\mu(-s, s)$ is linear and upper semicontinuous for weak convergence because $(-s, s)$ is open. By the localization theorem (Kannan et al., 1995; Fradelizi & Guédon, 2004), the supremum of this functional over the log-concave probability measures on K that satisfy $\int (g_x - p) d\mu \geq 0$ is attained at a point mass or at a law with log-affine density on a segment. Hence some member of $\mathcal{F}$ is feasible and has $\text{pr}(|W| < s) < q$, that is, $Q_q(|W|) \geq s$. Since $s < Q_q(|W|)$ was arbitrary the claim follows. Only continuity of the kernel and its decay at infinity were used, so the argument applies to the average kernel $\bar{g}$ of § 4.3. $\square$

For fixed slope and length, the translations c of a member of $\mathcal{F}$ that satisfy the constraint form an interval, because $c \mapsto \text{pr}(|W + c + e| \leq 1)$ is log-concave (Prékopa, 1973), and $c \mapsto \text{pr}(|W + c| \leq s)$ is log-concave, so $c \mapsto Q_q(|W + c|)$ is quasi-convex and is maximized at an end of that interval. Each end is found by a root-finding in c , and the quantile by a second root-finding, using (2). This two-dimensional computation over slope and length, and an independent differential-evolution search over the three-parameter family, agree to 1.5×10^{-13} at all 73 tabulated values of $x \geq 0.006$ (numerical).

Proof of Lemma 1. Exponential tails are log-concave, which gives one inequality. Conversely, let $Y \in \mathcal{L}$ with $\text{pr}(Y + Z \leq 0) \geq p$. If $Y = y_0$ is a point mass then $y_0 \leq -\Phi^{-1}(p) = \lim_{u \rightarrow \infty} \{b_u(p) - u^{-1} \log(1/q)\}$. Otherwise $F = F_Y$ is continuous and $\log F$ is concave (Prékopa, 1973). Let $v = F^{-1}(q)$ and $G(y) = F(v + y)$. The point 0 is interior to $\{G > 0\}$, so the right derivative $u \geq 0$ of $\log G$ at 0 is a supergradient, and $G(y) \leq qe^{uy}$ for all y . If $u = 0$ then $G \leq q < 1$, which is impossible, so $u > 0$ and $G(y) \leq \min(qe^{uy}, 1)$, which is the distribution function of $b_0 - E_u$ with $b_0 = u^{-1} \log(1/q)$. Hence $Y' = v + b_0 - E_u$ lies stochastically below Y and has q -quantile v , and

$$p \leq \text{pr}(Y + Z \leq 0) \leq \text{pr}(Y' + Z \leq 0) = m_u(v + b_0).$$

Since m_u is decreasing, $v \leq b_u(p) - b_0$. For monotonicity, let $q' > q$, let Y be feasible at (q', q') with $v = F_Y^{-1}(q')$, and put $\rho = (q' - q)/(1 - q)$. Then Y conditioned on $Y \geq F_Y^{-1}(\rho)$ is log-concave, satisfies

the constraint at level $(q' - \rho)/(1 - \rho) = q$, because the removed mass has noisy constraint value at most one, and has q -quantile v . Point masses have value $-\Phi^{-1}(q') < -\Phi^{-1}(q) \leq c_q$. Finally, $c_q > 0$ because $v_q(u) = u^{-1} \log\{e^{u^2/2}\Phi(b_u - u) + e^{ub_u}\Phi(-b_u)\} = u/2 + o(u)$ as $u \rightarrow 0$, where $b_u = b_u(q)$ and $ub_u \rightarrow \log(1/q)$. $\square$

Proof of Theorem 1, lower bound. Fix $\epsilon > 0$ and choose u with $v_q(u) > c_q - \epsilon$. For $p' \in [q, (1+q)/2]$, $b_u(p') \geq b_{\min} > -\infty$. Let $Y = b_u(p') - E_u$ and $W = 1 + x^{1/2}Y$. With $\tau = 2x^{-1/2}$,

$$\Pr(|W + x^{1/2}Z| \leq 1) = p' - \Pr(Y + Z < -\tau) \geq p' - \eta_x, \quad \eta_x = e^{-u(\tau + b_{\min})/2} + \Phi\{-(\tau + b_{\min})/2\},$$

by splitting the event $Y + Z < -\tau$ according to whether $Z < -(\tau + b_{\min})/2$. Take $p' = q + \eta_x$, which lies in $[q, (1+q)/2]$ for small x . Since $\Pr(|W| \leq s) \leq F_W(s)$, $Q_q(|W|) \geq 1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1} \log(1/q)\}$. As $\eta_x \rightarrow 0$ and b_u is continuous, $\liminf_{x \rightarrow 0}\{R_{q,q}(x) - 1\}/x^{1/2} \geq c_q - \epsilon$. $\square$

Proof of Proposition 3. $\bar{F}(T) < p_k$ exactly when $N = \#\{i : |V_i| < \bar{F}^{-1}(p_k)\} \geq k$. N is a sum of independent indicators whose success probabilities average p_k , and $k - 1 \geq Kp_k$. By Hoeffding (1956), $\Pr(N \leq k - 1)$ is smallest when all success probabilities are equal. Hence $\Pr(N \geq k) \leq \Pr\{\text{Bin}(K, p_k) \geq k\} = \Pr\{\text{Beta}(k, K + 1 - k) \leq p_k\} = \delta$. $\square$

The case $p > q$

In the proof of the upper bound of Theorem 1, replace $1 - q$ by $1 - p$ for the noisy budget, choose any split $\beta_L + \beta_R = 1 - q$ of the latent budget, and use $F_Y^{-1}(1 - \beta_R) \leq c_{1-\alpha_R, 1-\beta_R}$ at each end. Since $c_{p', q'}$ decreases in p' , the worst case has $\alpha_L + \alpha_R = 1 - p$, which gives $R_{p,q}(x) \leq 1 + C_{p,q}x^{1/2}$ for every x , with $C_{p,q}$ nonincreasing in p . The proof of the lower bound applies with $b_u(q)$ replaced by $b_u(p)$ and $p' = p + \eta_x$, giving $R_{p,q}(x) \geq 1 + c_{p,q}x^{1/2} - o(x^{1/2})$ for every $p \geq q$. This proves Corollary 1, whose numerical intervals are certified in § S3.

Bi-log-concave laws

Corollary S1. Let $\mathcal{B}$ be the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave (Dümbgen et al., 2017); $\mathcal{L} \subset \mathcal{B}$. Define $R_{p,q}^{\mathcal{B}}$ by (1) with $\mathcal{B}$ in place of $\mathcal{L}$. Then Lemma 1 holds over $\mathcal{B}$ with the same $c_{p,q}$, and for $p \geq q$ and every $x > 0$, $R_{p,q}^{\mathcal{B}}(x) \leq 1 + C_{p,q}x^{1/2}$, while $R_{q,q}^{\mathcal{B}}(x) = 1 + c_{q,q}x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$. In particular Corollary 1 holds over $\mathcal{B}$.

Proof. The proof of Lemma 1 uses only the concavity of $\log F_Y$ at the q -quantile, so its upper bound holds for every Y with $\log F_Y$ concave, and exponential tails are in $\mathcal{L}$, which gives equality. In the proof of Theorem 1, $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave, and $(-W - 1)/x^{1/2}$ has a concave log distribution function because $\log(1 - F_W)$ is concave and F_W is continuous. The lower bound uses laws in $\mathcal{L} \subset \mathcal{B}$. $\square$

The class $\mathcal{B}$ contains multimodal laws. Proposition 1, the certified values and Theorems 2 and 3 use log-concavity of the density, and Theorem 2 also uses that Gaussian convolution preserves the class, so they do not transfer. With homogeneous noise and $p_k - q \geq 10^{-3}$ at $q = 0.9$, the noisy threshold itself covers a bi-log-concave latent residual with probability at least q on the order-statistic event.

S2 The shape of the sharp radius

Figure S1 shows $R_{q,q}(x) - 1$ computed by grid maximization over the extremal family; grid values bound $R_{q,q}$ from below. The levels at which the grid value crosses 1 are $x^*(q) \approx 0.034, 0.017, 0.0062$ and 0.0012

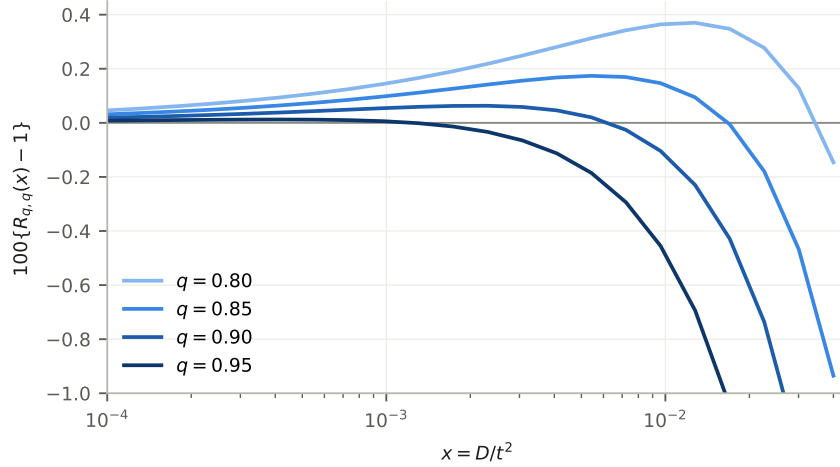


Figure S1: Sharp latent radius relative to the noisy threshold, $100\{R_{q,q}(x) - 1\}$, for four coverage levels, computed by grid maximization over the extremal family, which bounds $R_{q,q}$ from below.

for $q = 0.8, 0.85, 0.9$ and 0.95 , and the largest widening found is 0.37%, 0.17%, 0.063% and 0.012%. Below $x^*(q)$ the noisy threshold must be widened; above it, grid values below 1 do not by themselves show that shrinking is safe, which the certified upper bounds of § S3 do. The exact grid values lie below $1 + c_{0.9}x^{1/2}$ at $x = 10^{-6}, \dots, 10^{-2}$, as Theorem 1 requires.

Centred laws and the transition

The following values are computed in double precision by `experiments/e32_centering.py` and are not certified. For $q = 0.8, 0.9$ and 0.95 : $\kappa_q^* = 6.8118, 20.1823$ and 50.0741 , attained at $u^* = 0.11328, 0.044286$ and 0.018943 . The two upper bounds in Fig. 1 cross at $\kappa = 10.829, 26.221$ and 59.466 , where L_q/U_q is smallest, 0.770, 0.879 and 0.927; as $\kappa \rightarrow \infty$ the ratio tends to $1 - \log(1/q) = 0.777, 0.895$ and 0.949 . At $q = 0.9$, $L_q(25) = 0.017393$ and $L_q(50) = 0.0089448$. For the mean-zero laws of Proposition 2, the coefficient with $\beta = 0.2$ and $\ell = 2$ at $q = 0.9$ is 0.0156540; the expansion is accurate to 6×10^{-5} at $x = 1.3 \times 10^{-4}$ but fails at $x \approx 1.1 \times 10^{-3}$, where the noise reaches the end of the support and the threshold may already be shrunk. The bound of Theorem 2, including its form with $|E(W)| \leq B$, was checked on 642 laws from the family of Proposition 2, with no violation.

The limit in Theorem 3 is taken with κ fixed. Applying the small- u construction to a centred law, that is, $\kappa = x^{-1/2}$, gives in the limit a left exponential tail of rate $1 - \log(1/q)$ whose noise-free central coverage is $q\{1 - e^{-2+2\log(1/q)}\}$, 0.750 at $q = 0.9$, far below the constraint. So $\{1 - \log(1/q)\}/2$ is not a lower bound for the centred coefficient.

S3 Certification

Box bounds for the radius

Write a member of $\mathcal{F}$ with $\beta \geq 0$ as $W = b - Y$, where Y has density proportional to $e^{-\beta y}$ on $[0, \ell]$. The ends of the family are included: $\ell = 0$ or $\beta = \infty$ gives a point mass and $\ell = \infty$ an exponential tail. For slopes $\beta' > \beta$ the likelihood ratio $e^{-(\beta' - \beta)y}$ is decreasing, so $Y_{\beta'}$ lies below Y_{β} in likelihood-ratio order, and for lengths $\ell < \ell'$ truncation gives $F_{\ell} \geq F_{\ell'}$. Thus W is stochastically increasing in β and b and decreasing

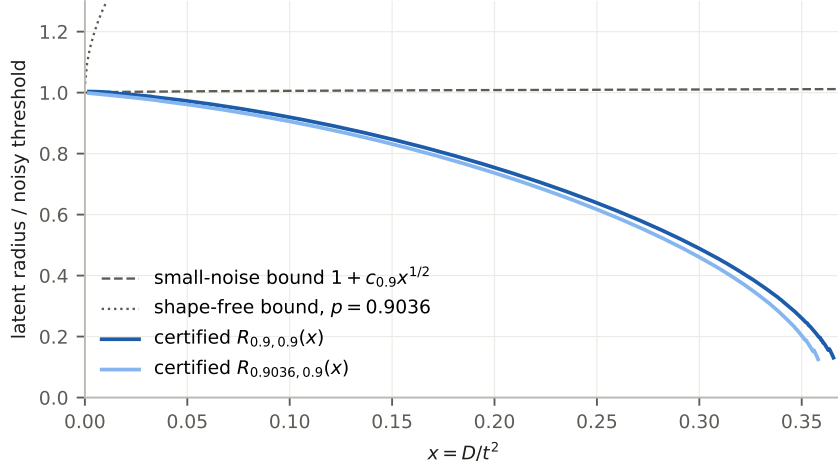


Figure S2: Certified sharp radius $R_{p,0.9}(x)$ for $p = 0.9$ and for the order-statistic level $p = 0.9036$, with the small-noise bound of Theorem 1 and the shape-free bound $1 + x^{1/2}z_{1-(p-q)/2}$, which follows from $|W| \leq |V| + |e|$.

in ℓ . On a box of parameters let W_{lo} and W_{hi} be the stochastically smallest and largest corners. Since $\text{pr}(W + e \leq \pm 1)$ and $\text{pr}(W \leq y)$ are expectations of decreasing functions of W , every law in the box satisfies

$$\text{pr}(|W + e| \leq 1) \leq \text{pr}(W_{\text{lo}} + e \leq 1) - \text{pr}(W_{\text{hi}} + e \leq -1), \quad \text{pr}(|W| \leq s) \geq \text{pr}(W_{\text{hi}} \leq s) - \text{pr}(W_{\text{lo}} \leq -s).$$

The density bound $\text{pr}(|W + e| \leq 1) \leq 2 \sup f_W = 2\beta/(1 - e^{-\beta\ell})$ is also available. A box whose first bound is below p , or whose second bound is at least q , contains no law that violates $R_{p,q}(x) \leq s$, and a branch and bound over boxes certifies $R_{p,q}(x) \leq s$. The location is confined to $[-1 + x^{1/2}\Phi^{-1}(p), b_{\max}]$: the lower end holds because $W \leq b$; for the upper end take $k = 7x^{1/2}$; the density of W increases towards b , so $\text{pr}(|W| \leq 1 + k) \leq r/(1 + r)$ with $r = (2 + 2k)/(b - 1 - k)$, and $\text{pr}(|e| > k) = 2\Phi(-7)$. The parameters $\beta, \ell \in [0, \infty]$ are compactified.

Proposition S1. For all $x, h \geq 0$, $R_{p,q}(x + h) \leq R_{p,q}(x) + c_q h^{1/2}$.

Proof. If W is feasible at noise level $x + h$, then $W + h^{1/2}Z'$, with $Z' \sim N(0, 1)$ independent of (W, Z) , is log-concave and feasible at level x . So $s = Q_q(|W + h^{1/2}Z'|) \leq R_{p,q}(x)$, and Theorem 1 at scale s gives $Q_q(|W|) \leq s + c_q h^{1/2}$. $\square$

We certified $R_{0.9,0.9}$ and $R_{0.9036,0.9}$ on $0.002 \leq x \leq 0.366$ with step 5×10^{-4} , using Proposition S1 between grid points. The closed forms agree with high-precision quadrature to 10^{-11} and boxes are cleared with a margin of 10^{-9} , so the values are certified up to floating-point evaluation of the closed forms; this is not interval arithmetic. For $0.05 \leq x \leq 0.3$ the certified bound exceeds a feasible value by at most a relative 3×10^{-4} . The certificate fails only just before the feasibility edges $x \approx 0.370$ for $p = 0.9$ and 0.361 for $p = 0.9036$. Below $x = 0.01$ the lookup also uses Theorem 1 and its $p \geq q$ form, which are within 0.3% of a feasible value there (Fig. S2). At $q = 0.9$ the certified bounds fall below 1 from $x = 0.013$ on and reach 0.973 at $x = 0.05$.

Ball-arithmetic certificates for the constants

Every deciding inequality is evaluated in Arb ball arithmetic; floating point only proposes points, balls are converted to doubles with outward rounding, and sums of bounds are formed as balls.

1. $m_u(b)$ decreases in b , and in u because E_u is stochastically decreasing in its rate. Hence $b_u(p)$ is nonincreasing in u , and on $[u_0, u_1]$, $b_u(p) - u^{-1} \log(1/q) \leq b_{u_0}(p) - u_1^{-1} \log(1/q)$. A ball with $m_u(b) < p$ proves $b_u(p) < b$.
2. For $u \geq u_{\max}$ the value is at most $b_{u_{\max}}(p)$. For small u , let $b(u) = \{\log(1/p) + u^2/2 - \log(1 - u^3)\}/u$. Then $e^{-ub+u^2/2} = p(1 - u^3)$, and the Mills ratio gives $\Phi(-b) \leq pu^3$ whenever $\phi\{\log(1/p)/u\} \leq pu^2 \log(1/p)$. This holds on $(0, u_{\min}]$ once it holds at $u_{\min} \leq \log(1/p)/2^{1/2}$, because the left side divided by u^2 increases there. So $b_u(p) \leq b(u)$, and for $p \geq q$ the resulting bound increases in u .
3. By Lemma 1, $c_{p',q'} \leq \gamma$ if and only if $1 - q' \geq \varphi_\gamma(1 - p')$, where $\varphi_\gamma(a) = \max[0, \sup_u 1 - \exp\{-u(b_u(1 - a) - \gamma)\}]$. Hence $C_{p,q} \leq \gamma$ if and only if $\varphi_\gamma(a_L) + \varphi_\gamma(a_R) \leq 1 - q$ whenever $a_L + a_R = 1 - p$. Since φ_γ is nondecreasing, it suffices to check this at the ends of an adaptive partition of a_L , with φ_γ bounded above by the same monotone branch and bound in u .

The certificates give the enclosures of c_q in § 3, $-0.0571845 \leq c_{0.9036,0.9} \leq C_{0.9036,0.9} \leq -0.057$, and $C_{q+\delta,q} \leq 0$ for $(q, \delta) = (0.8, 0.005)$, $(0.9, 0.001)$ and $(0.95, 0.0002)$. Lower bounds $c_{q+\delta,q} > 0$ at $\delta = 0.0043$, 0.00079 and 0.00015 come from single points u , and $C_{p,q} \geq c_{p,q}$. Double-precision root finding puts the smallest slack at 4.4×10^{-3} , 8.0×10^{-4} and 1.6×10^{-4} .

S4 Heterogeneous and estimated noise variances

Heterogeneous known variances

In the heterogeneous rule, ε bounds, for each data set, the sup-norm error of the average kernel after compressing the D_i to 32 support points: the maximum over a grid of spacing $h = 10^{-3}$ plus $Mh^2/8$, with M a bound on the second derivative; it was about 2×10^{-4} . The radius R^{mix} is certified for each data set by the box bounds, which hold term by term for a mixture kernel, and the shape-free bound is used if the certificate fails.

Replacing the average kernel by the kernel of a single variance is not safe. With the mean variance plugged in, the radius is 1.4% too short at mean $x = 0.2$ with lognormal spread 1 (numerical). The smallest variance also fails: take $W = b - E_2$ with $b = 1.0435647$ and noise an equal mixture of $N(0, 10^{-8})$ and $N(0, 5 \times 10^{-4})$. The noisy coverage of $[-1, 1]$ exceeds 0.9, while the radius $1 + c_{0.9}10^{-4}$ given by the smaller variance covers W with probability 0.89977; this was checked in 192-bit ball arithmetic.

Estimated variances

Suppose the D_i are unknown and estimates $\hat{D}_i$ are available that are independent of the scores, as for the sample means and variances of normal unit data. The order-statistic event of Proposition 3 involves the scores only. Let $\mathcal{E}_\eta$ be an event determined by the $\hat{D}_i$ with $\text{pr}(\mathcal{E}_\eta) \geq 1 - \eta$.

Lemma S1. *Suppose that on $\mathcal{E}_\eta$ a continuous kernel u_t , vanishing at $\pm\infty$, satisfies $u_t \geq \bar{g}_{D,t}$ pointwise for every $t > 0$. Then $s = T R_{p_k,q}[u_T]$, the radius (1) with kernel u_T , covers a new latent residual with probability at least q , conditionally on the data, with probability at least $1 - \delta - \eta$. If $\mathcal{D}_\eta$ is a confidence set with $\text{pr}(D \in \mathcal{D}_\eta) \geq 1 - \eta$, the same holds for $s = T \sup_{d \in \mathcal{D}_\eta} R_{p_k,q}^{\text{mix}}(d/T^2)$.*

Proof. On both events, $\int u_T dG \geq \int \bar{g}_{D,T} dG \geq p_k$, and Proposition 1 applies to u_T . The union bound gives $1 - \delta - \eta$, with no independence between the two events needed. $\square$

Lemma S2. *For $\lambda \geq 1$, $R_{p,q}^{\text{mix}}(\lambda x_1, \dots, \lambda x_K) \leq \lambda^{1/2} R_{p,q}^{\text{mix}}(x_1, \dots, x_K)$.*

Proof. If W is feasible at λx , then $U = \lambda^{-1/2}W$ satisfies the constraint at threshold $\lambda^{-1/2} \leq 1$, hence at threshold 1, so $Q_q(|W|) = \lambda^{1/2}Q_q(|U|) \leq \lambda^{1/2}R_{p,q}^{\text{mix}}(x)$. $\square$

A confidence box for all K variances needs level $1 - \eta/K$ per area and is useless with one or two degrees of freedom. Survey practice instead models variances through a few parameters, as generalized variance functions do. In the simplest such model $D_i = \sigma^2 a_i$ with known a_i , for instance $a_i = 1/n_i$, and the pooled estimate satisfies $\nu \hat{\sigma}^2 / \sigma^2 \sim \chi_\nu^2$ with $\nu = \sum_i (n_i - 1)$. The confidence set is the segment $\{\sigma^2 a : \ell \leq \sigma^2 \leq h\}$. Lemma S2 covers the supremum over it by finitely many points without assuming that R is monotone in σ^2 : for $\ell = s_0 < \dots < s_m = h$,

$$\sup_{\ell \leq \sigma^2 \leq h} R_{p_k,q}^{\text{mix}}(\sigma^2 a / T^2) \leq \max_j (s_{j+1}/s_j)^{1/2} U_j, \quad U_j \geq R_{p_k,q}^{\text{mix}}(s_j a / T^2).$$

Plugging in the lower end ℓ alone would not be justified, since R can increase with the noise level (Theorem 1).

In simulations, areas have n_i normal units, so $D_i = \sigma_i^2/n_i$ and the within-area variances are independent of the direct estimates; $K = 110$, mean $D = 0.577$, and 100 replications per setting and law (normal, Laplace, truncated Laplace). The rule uses $\delta = 0.04$ and $\eta = 0.01$, split as 0.009 below and 0.001 above the pooled interval for σ^2 , and needs 18 grid points on average. The radii below use converged grid values at those points; for the first 30 data sets of every setting and law (270 in all) the rule was rerun with certified upper bounds U_j at every grid point, as Lemma S1 requires. Every data set was certified, and the half-width grew by 0.14% on average and by at most 1.0%, at about 3.6 minutes per data set on one core.

1. With $n_i \in \{2, 3\}$, $\nu \approx 165$, the rule is 6.3–6.5% shorter than noisy conformal calibration and 2.9–3.0% longer than the known-variance rule at the same level, which is 9.0–9.1% shorter than noisy conformal calibration. With $n_i \in \{5, \dots, 10\}$, $\nu \approx 740$, the figures are 7.1–7.7% and 1.6–1.7%. Relative to the known-variance rule at $\delta = 0.05$, which needs no budget for the variances and meets the same final target, 62–63% and 73–74% of its gain survive. Reliability was at least 0.98.
2. The supremum over the interval exceeded the value at its lower end by 0.2%.
3. When the within-area variances differ, with lognormal spread 0.5, the common-variance model is wrong and no guarantee applies; the rule was still reliable, at least 0.99, and 5.9–6.5% shorter than noisy conformal calibration.
4. Without pooling, with each area's variance estimated from ν own degrees of freedom, a kernel envelope built from area-wise intervals and a binomial bound on the number of misses is 31–40% longer than noisy conformal calibration at $\nu = 2$, within 1.3% of it at $\nu = 50$, and 4.7–5.0% and 6.6–7.2% shorter at $\nu = 200$ and 1000 (30 replications per law; grid values). Whether a better area-wise construction exists is open.

S5 The shape-free comparison rule

Proposition S2. Let $\bar{g}$ be a kernel that is symmetric, continuous and strictly decreasing in $|w|$, with $m = \bar{g}(0)$, and suppose that $qm < p < m$. Over all laws G on the real line with $\int \bar{g} dG \geq p$, the supremum of $Q_q(|W|)$ is the radius r that solves $\bar{g}(r) = (p - qm)/(1 - q)$.

Proof. Let $\tau = (p - qm)/(1 - q) \in (0, m)$. Since $m - \bar{g}(W) \geq 0$, Markov's inequality gives $\text{pr}\{\bar{g}(W) < \tau\} \leq \{m - p\}/(m - \tau) = 1 - q$, and $\bar{g}(W) < \tau$ exactly when $|W| > r$, so $Q_q(|W|) \leq r$. Conversely, for small $\epsilon > 0$ put mass $q - \epsilon$ at 0 and mass $1 - q + \epsilon$ at the point $w_\epsilon > 0$ with $\bar{g}(w_\epsilon) = \{p - (q - \epsilon)m\}/(1 - q + \epsilon)$. This law meets the constraint with equality, has $Q_q(|W|) = w_\epsilon$, and $w_\epsilon \rightarrow r$ as $\epsilon \rightarrow 0$. $\square$

The average kernel of § 4.3 satisfies these conditions, so for a fixed rank the shape-free rule of Table 1 is the smallest radius that the single order-statistic constraint allows without a shape assumption. With 1 in place of $\bar{g}(0)$ it contains the bound of Guille-Escuret & Ndiaye (2024). It is not a bound on procedures that use several thresholds or tune the rank on independent data, as LatentCP (Zheng et al., 2026) can.

S6 Simulation details and an application

Table 1

Conditional coverage is computed from closed-form latent distribution functions. Each rule’s reliability has Monte Carlo standard error at most 0.018 with 150 data sets; for instance $145/150 = 0.967$ has the 95% binomial interval $[0.924, 0.989]$. Both log-concave rules use the order-statistic level $p_k = 0.9068$ of $\delta = 0.05$. Every heterogeneous radius was certified, at relative tolerance 10^{-3} in 368 of 600 data sets, 3×10^{-3} in 222 and 10^{-2} in 10. The order-statistic level accounts for most of the gain over the $p = q$ table, 3.0% of width, of which 1.2% comes from raising p from 0.9036 to 0.9068. The shape-free rule was run at every fixed rank 106–110 and all results are stored; rank 107 is reported because it is best on average, which favours the comparison rule. The Fay–Herriot interval falls short of the target under the Laplace law by about 1.3 standard errors.

Oracle-matched widths

In a larger experiment with seven latent shapes, common and heterogeneous noise and 2000 data sets per setting, each rule’s width was rescaled by a single factor, chosen with knowledge of the latent law, to reach reliability 0.95 exactly. At matched reliability noisy conformal calibration was shortest or tied, the Fay–Herriot interval was 0–11% longer and the log-concave rule 1–8% longer. The log-concave rule with its guarantee was about 15% wider than its own oracle-matched width, and the Fay–Herriot tolerance interval about 6%. So the advantage of the log-concave rule over noisy conformal calibration lies in obtaining a latent guarantee cheaply, not in a better interval shape.

School districts

In the finite population `apipop` (Lumley, 2004), 110 of 325 school districts are used for training and 110 for calibration, with 2 or 3 schools sampled per district. Coverage is the share of the remaining 105 districts that are covered. Design variances are estimated with 1–2 degrees of freedom and plugged in as known by every rule, a case that no guarantee covers. Over four conditions of 120 replications, the reliability of the heterogeneous rule was 0.942–0.992, of the mean-variance rule 0.983–1.000 and of the Fay–Herriot interval 0.908–0.992, below target in two conditions. The heterogeneous rule at level p_k was 0.8–5.2% shorter than the mean-variance rule with the $p = q$ table, mostly because of the level, and 10–16% longer than the Fay–Herriot interval.

S7 Reproducibility

All computations are in the public repository <https://github.com/kunhailP/unsampled-area-intervals>. The core routines are in `src/uai`: closed forms and the transition coefficient in `extremal.py`, the box certificates in `certify.py`, ball arithmetic in `interval.py`, and the procedures in `procedures.py` and `estimated.py`. The test suite, run by `make test`, checks the constants, the counterexamples and the centring bounds. The experiments are numbered scripts in `experiments`; `e27` produces the ball-arithmetic

certificates, e24 and e25 the certified tables, e20 Table 1, e29 and e30 the estimated-variance results, e31 the shape-free rule, e32 the centring and transition values, and fig_transition.py Fig. 1.

References

- DÜMBGEN, L., KOLESNYK, P. & WILKE, R. A. (2017). Bi-log-concave distribution functions. *J. Statist. Plan. Infer.* **184**, 1–17.
- FRADELIZI, M. & GUÉDON, O. (2004). The extreme points of subsets of s -concave probabilities and a geometric localization theorem. *Discrete Comput. Geom.* **31**, 327–35.
- GUILLE-ESCURET, C. & NDIAYE, E. (2024). From conformal predictions to confidence regions. *arXiv: 2405.18601*.
- HOEFFDING, W. (1956). On the distribution of the number of successes in independent trials. *Ann. Math. Statist.* **27**, 713–21.
- KANNAN, R., LOVÁSZ, L. & SIMONOVITS, M. (1995). Isoperimetric problems for convex bodies and a localization lemma. *Discrete Comput. Geom.* **13**, 541–59.
- LUMLEY, T. (2004). Analysis of complex survey samples. *J. Statist. Software* **9**, 1–19.
- PRÉKOPA, A. (1973). On logarithmic concave measures and functions. *Acta Sci. Math. (Szeged)* **34**, 335–43.
- ZHENG, M., ZHOU, W. & ZHU, S. (2026). Beyond prediction: conformal inference for latent distributional parameters. *arXiv: 2608.03607*.